



US012415629B1

(12) **United States Patent**
Jain et al.

(10) **Patent No.:** **US 12,415,629 B1**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **SYSTEM AND METHOD OF EJECTING A PAYLOAD FROM AN EXTRATERRESTRIAL SURFACE**

(71) Applicant: **Honeybee Robotics, LLC**, Longmont, CO (US)

(72) Inventors: **Anchal Jain**, Los Angeles, CA (US); **Sean O'Brien**, Kent, WA (US); **Dara Sabahi**, Longmont, CO (US); **Dean Bergman**, Altadena, CA (US)

(73) Assignee: **Honeybee Robotics, LLC**, Longmont, CO (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **19/006,885**

(22) Filed: **Dec. 31, 2024**

(51) **Int. Cl.**
B64G 1/64 (2006.01)

(52) **U.S. Cl.**
CPC **B64G 1/6457** (2023.08)

(58) **Field of Classification Search**
CPC B64G 1/6457; B64G 1/641; F41B 7/00; Y10T 292/0838

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,067,308 A * 1/1978 Andersen F41B 7/00 446/236
5,226,617 A * 7/1993 Panin B64G 1/641 292/35

FOREIGN PATENT DOCUMENTS

CN 102700729 A 10/2012
CN 110239966 A 9/2019
KR 10-2007-0113934 A 11/2007
RU 2779426 C1 9/2022

* cited by examiner

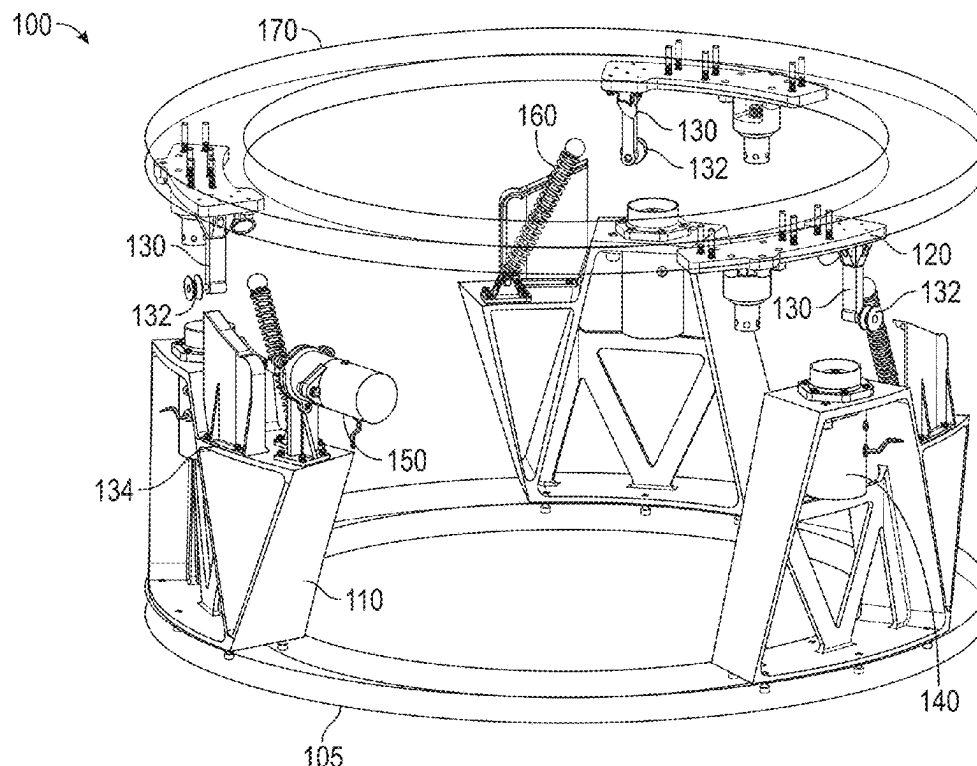
Primary Examiner — Valentina Xavier

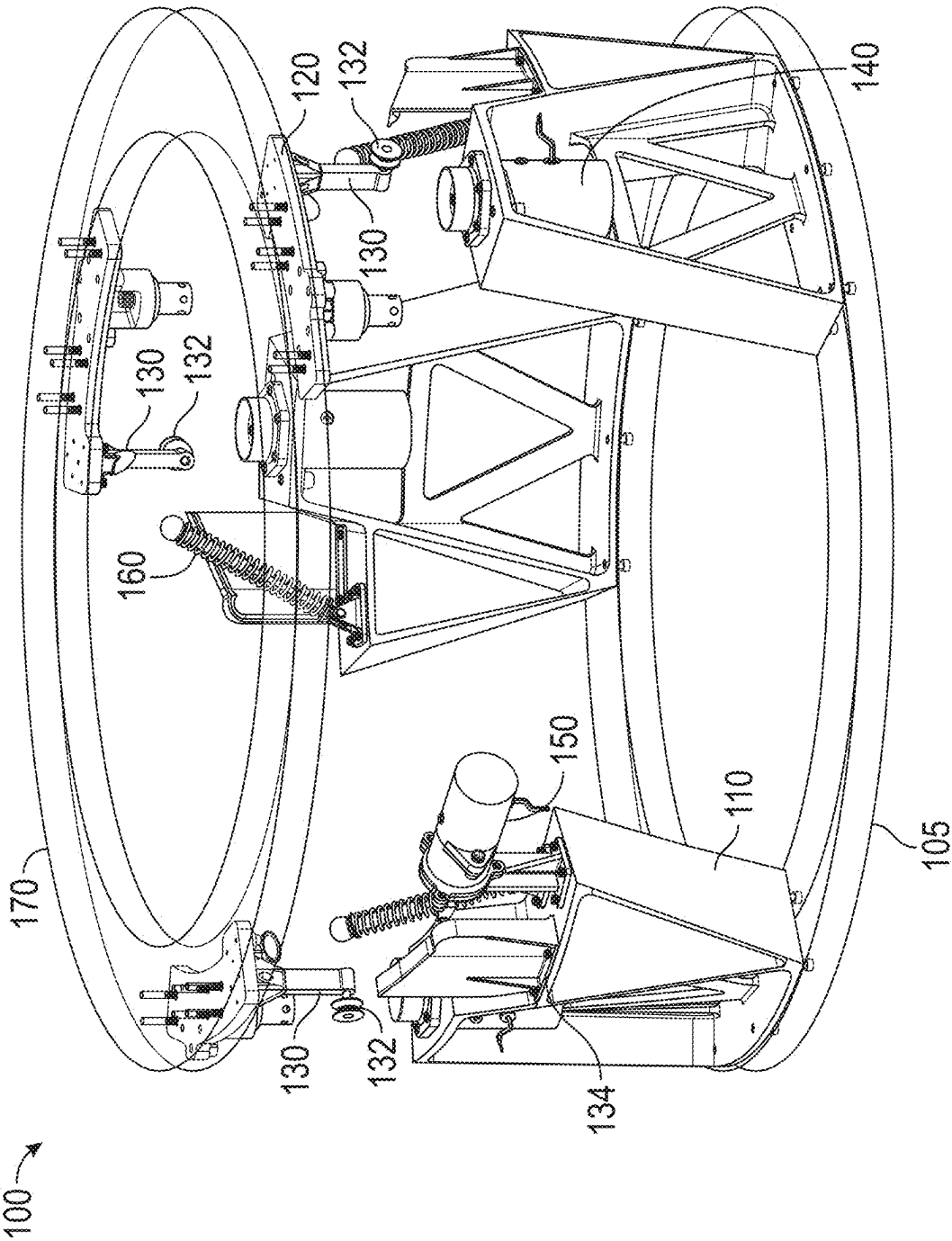
(74) *Attorney, Agent, or Firm* — McCarter & English, LLP; Dave S. Christensen

(57) **ABSTRACT**

A spin ejection mechanism and a method of launching objects from extraterrestrial sites is disclosed. The mechanism includes a first bracket. At least one second bracket is spaced apart from the first bracket. At least one guide mechanism is operably coupled between the first bracket and one of the at least one second brackets. The at least one guide mechanism is configured to axially restrain the second bracket. At least one releasable lock mechanism is operably coupled to the at least one second bracket and configured to initiate restrained movement of one of the at least one guide mechanism. A spring-loaded lock mechanism is coupled to the first bracket and configured to initiate unrestrained movement of the at least one guide mechanism.

18 Claims, 17 Drawing Sheets





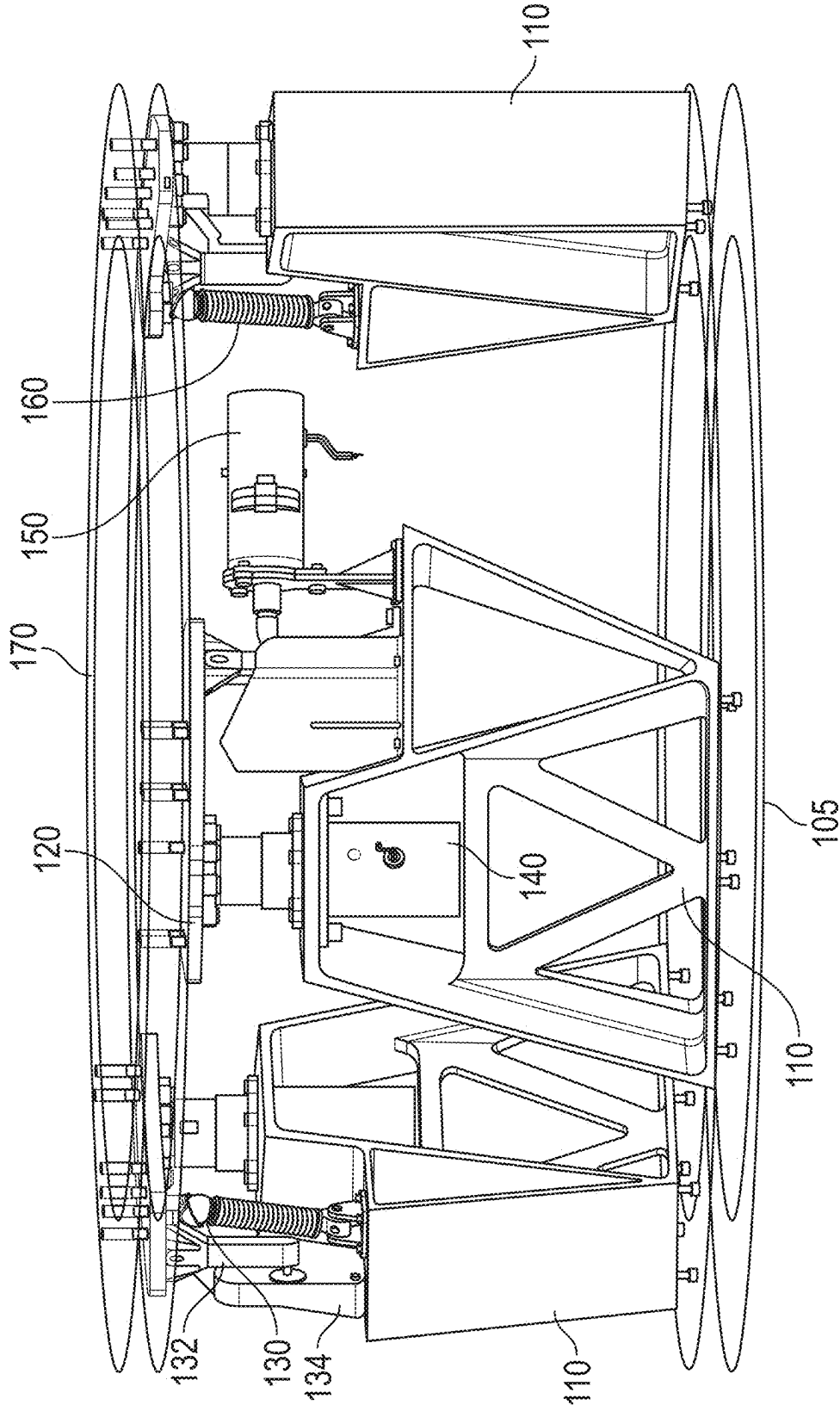


FIG. 2A

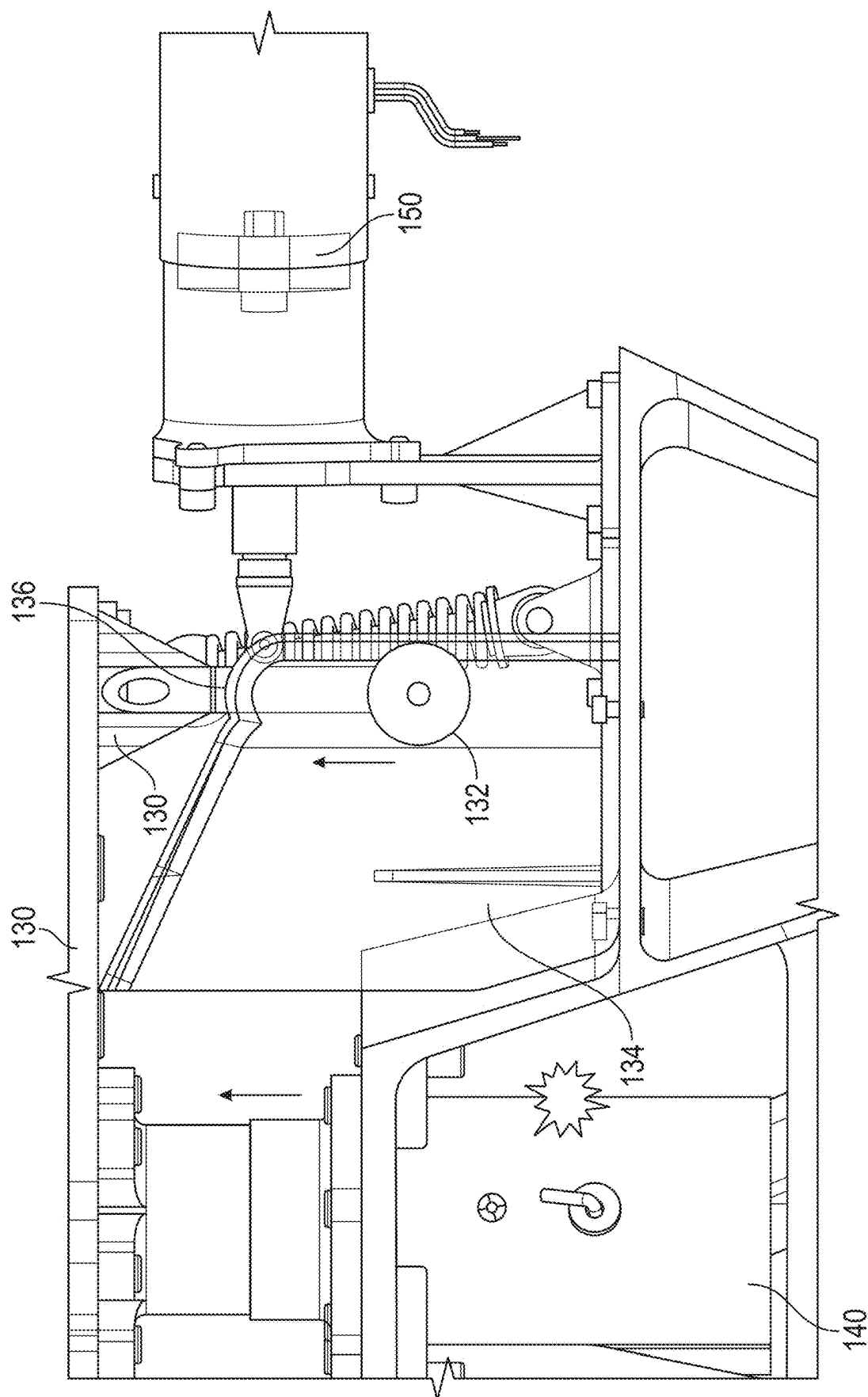


FIG. 2B

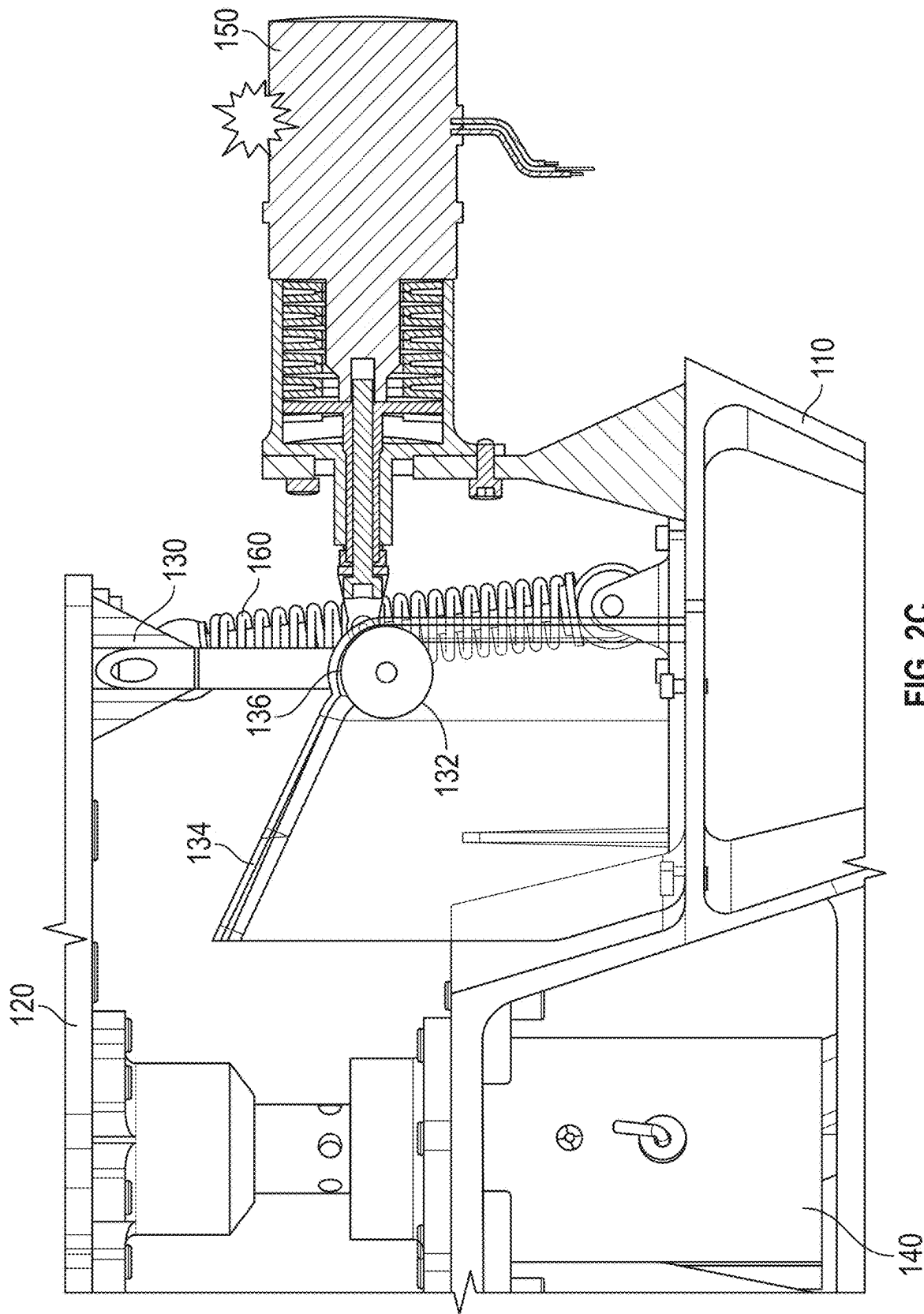


FIG. 2C

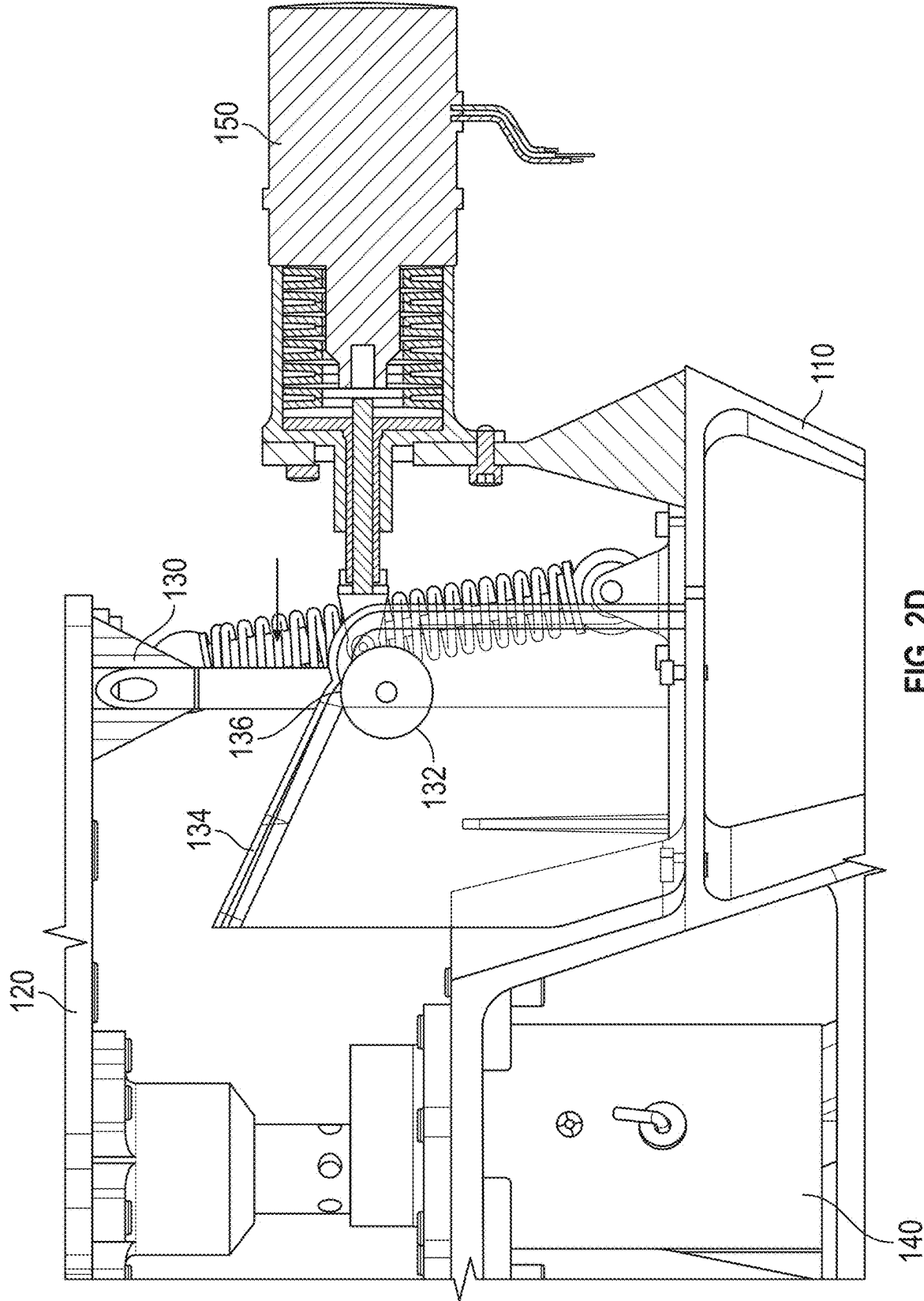


FIG. 2D

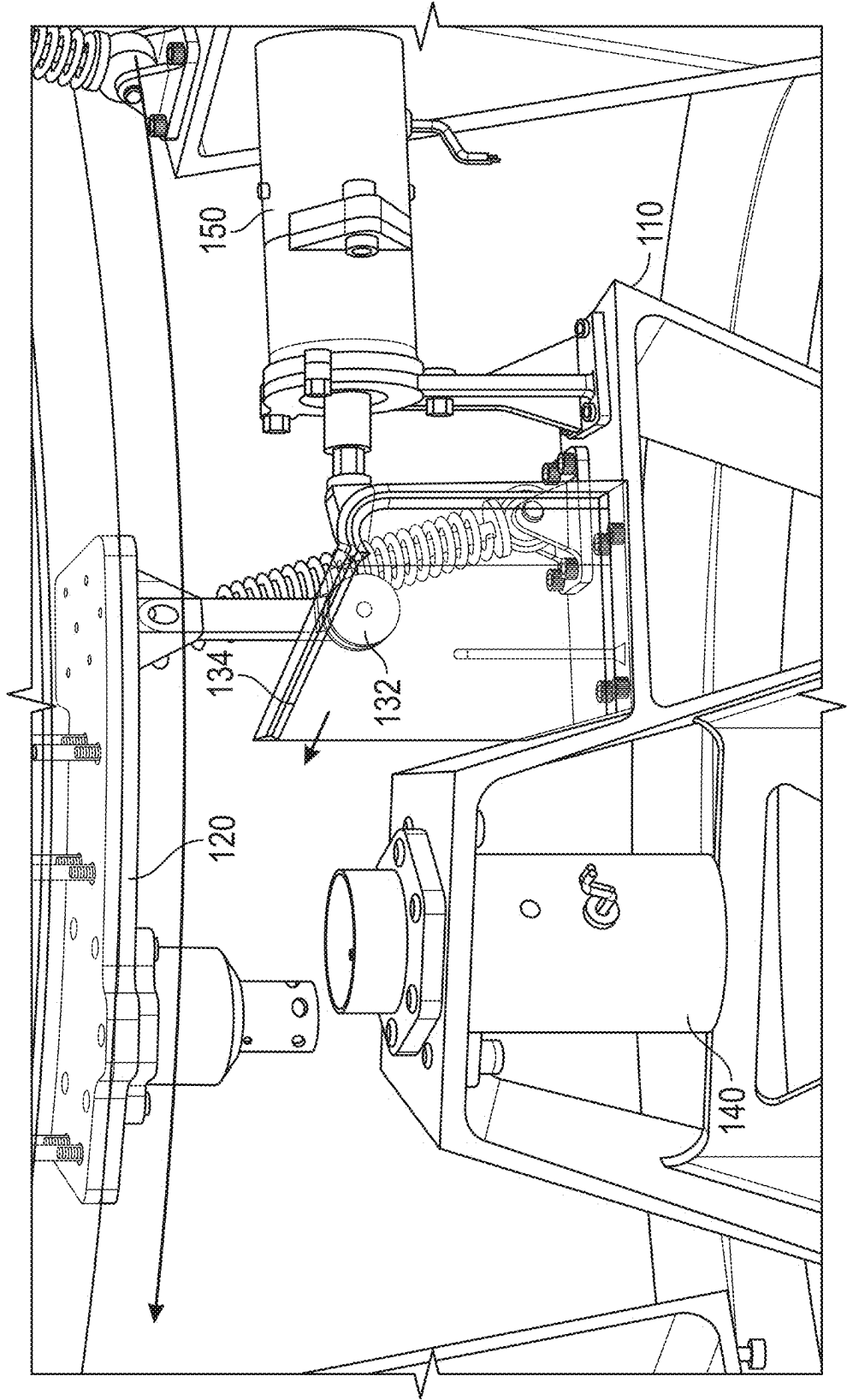


FIG. 2E

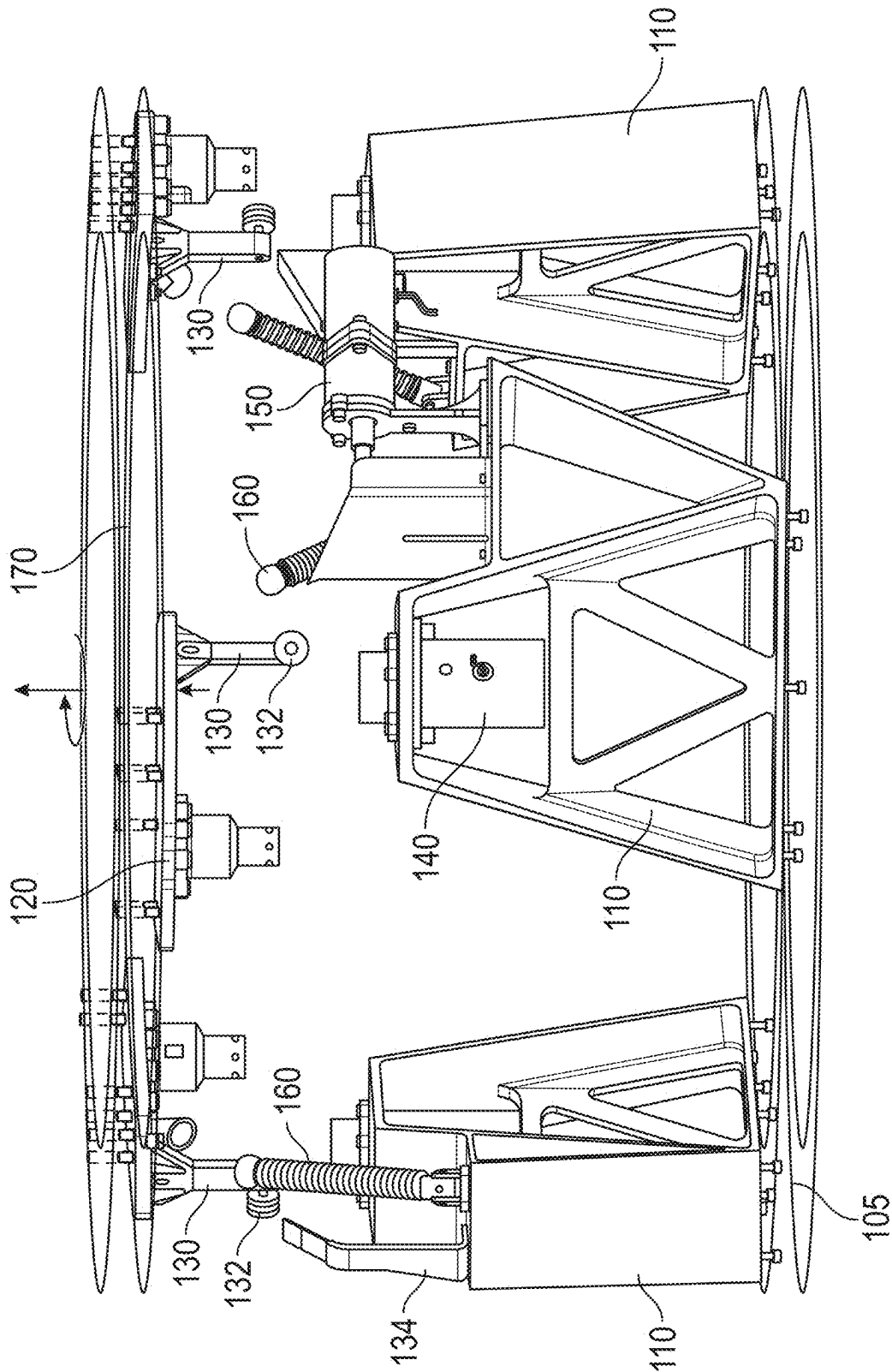


FIG. 2F

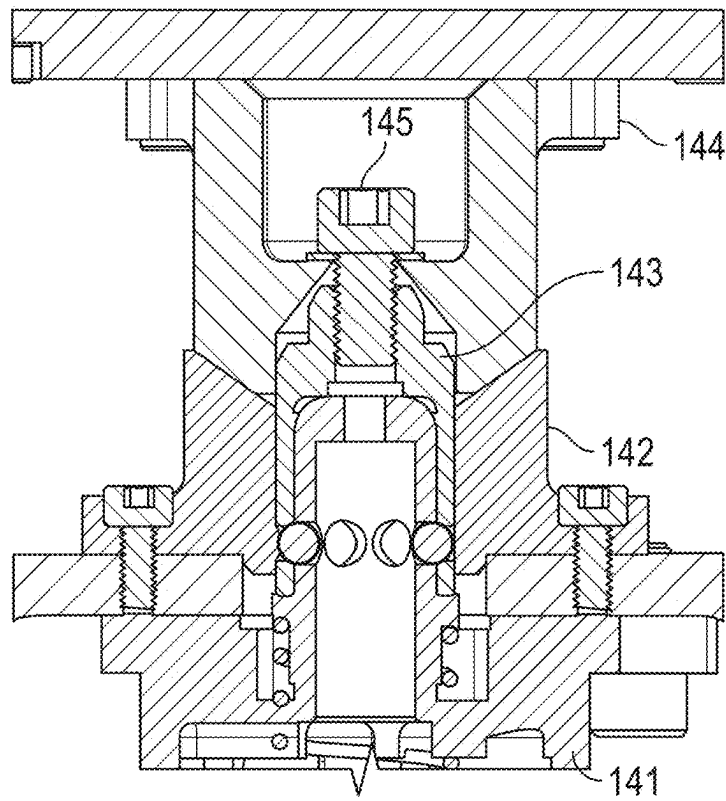


FIG. 3A

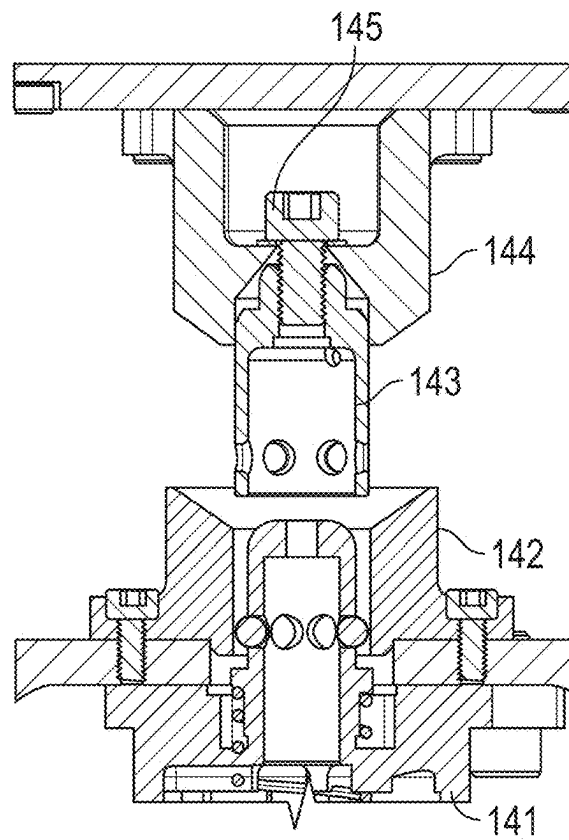


FIG. 3B

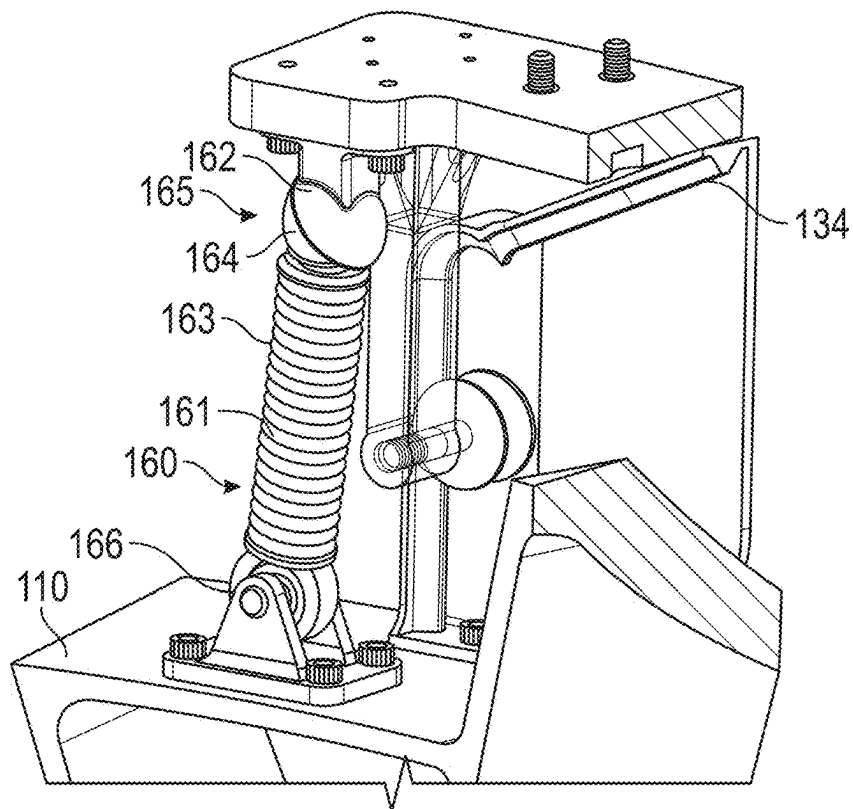


FIG. 4

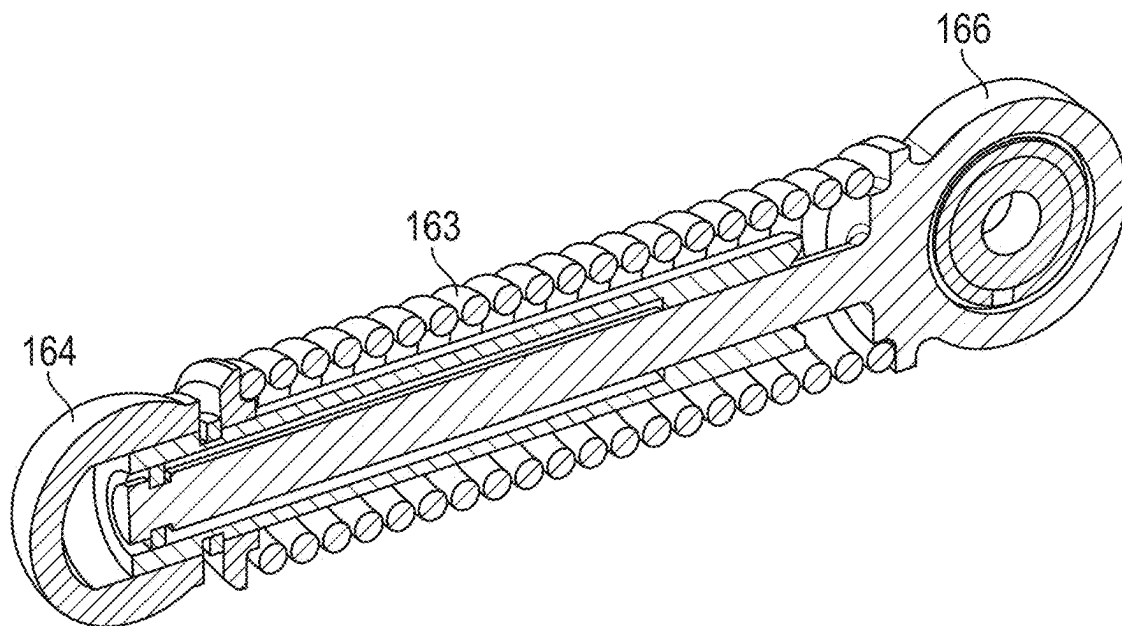


FIG. 5

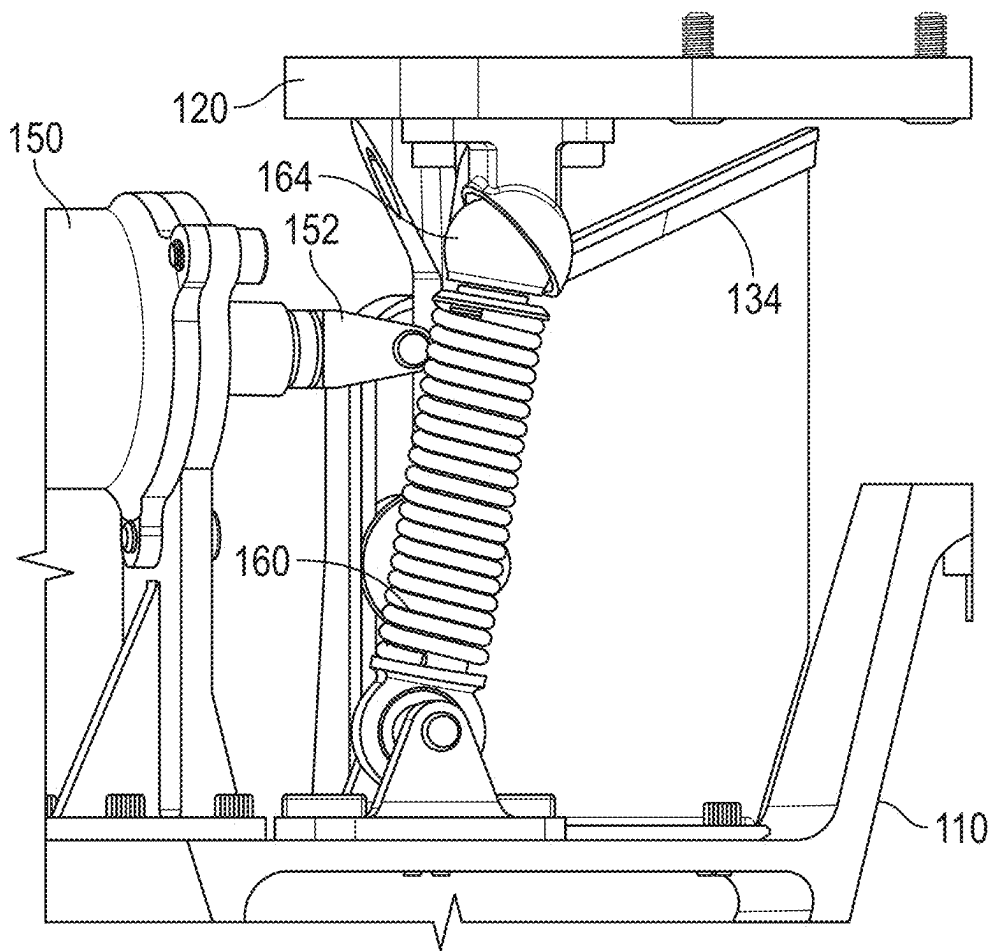


FIG. 6

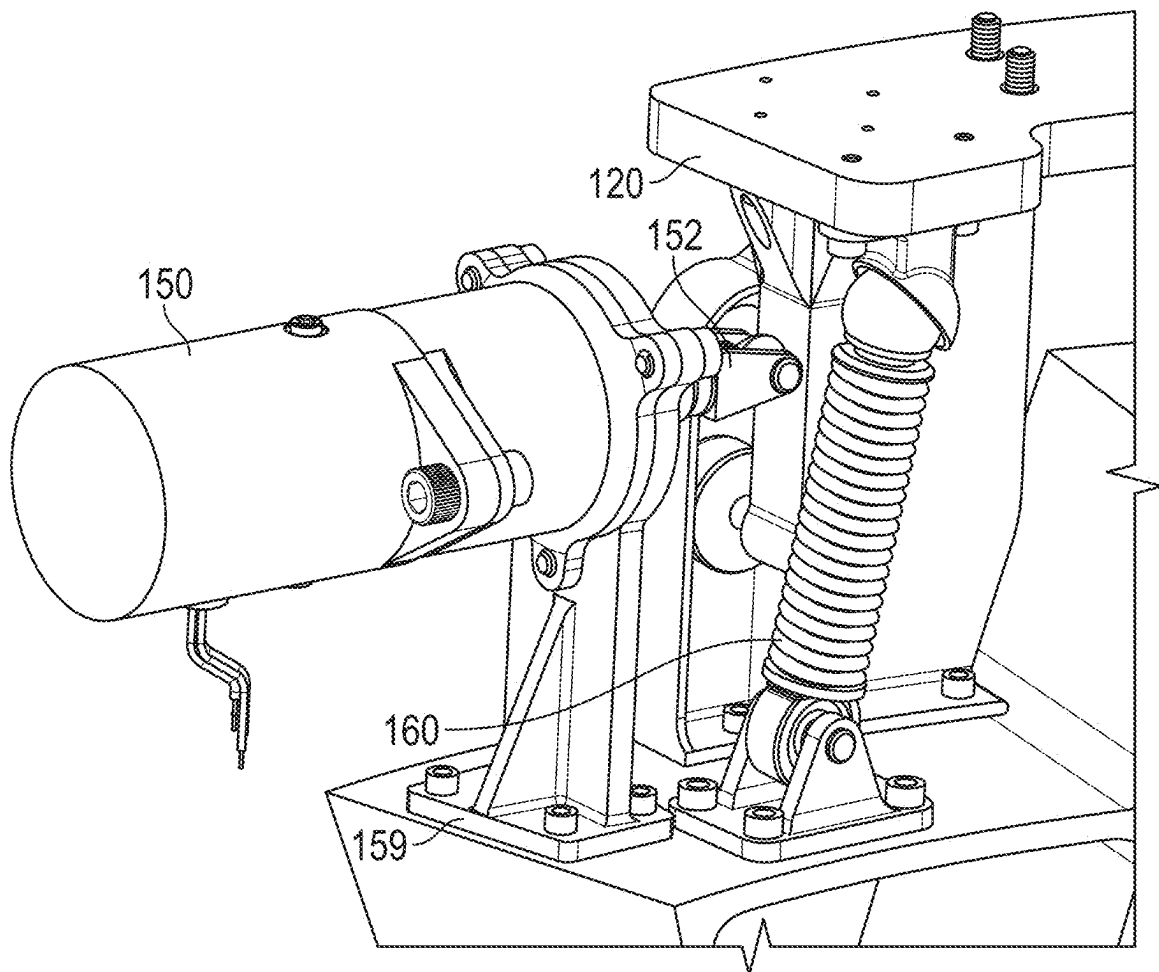


FIG. 7

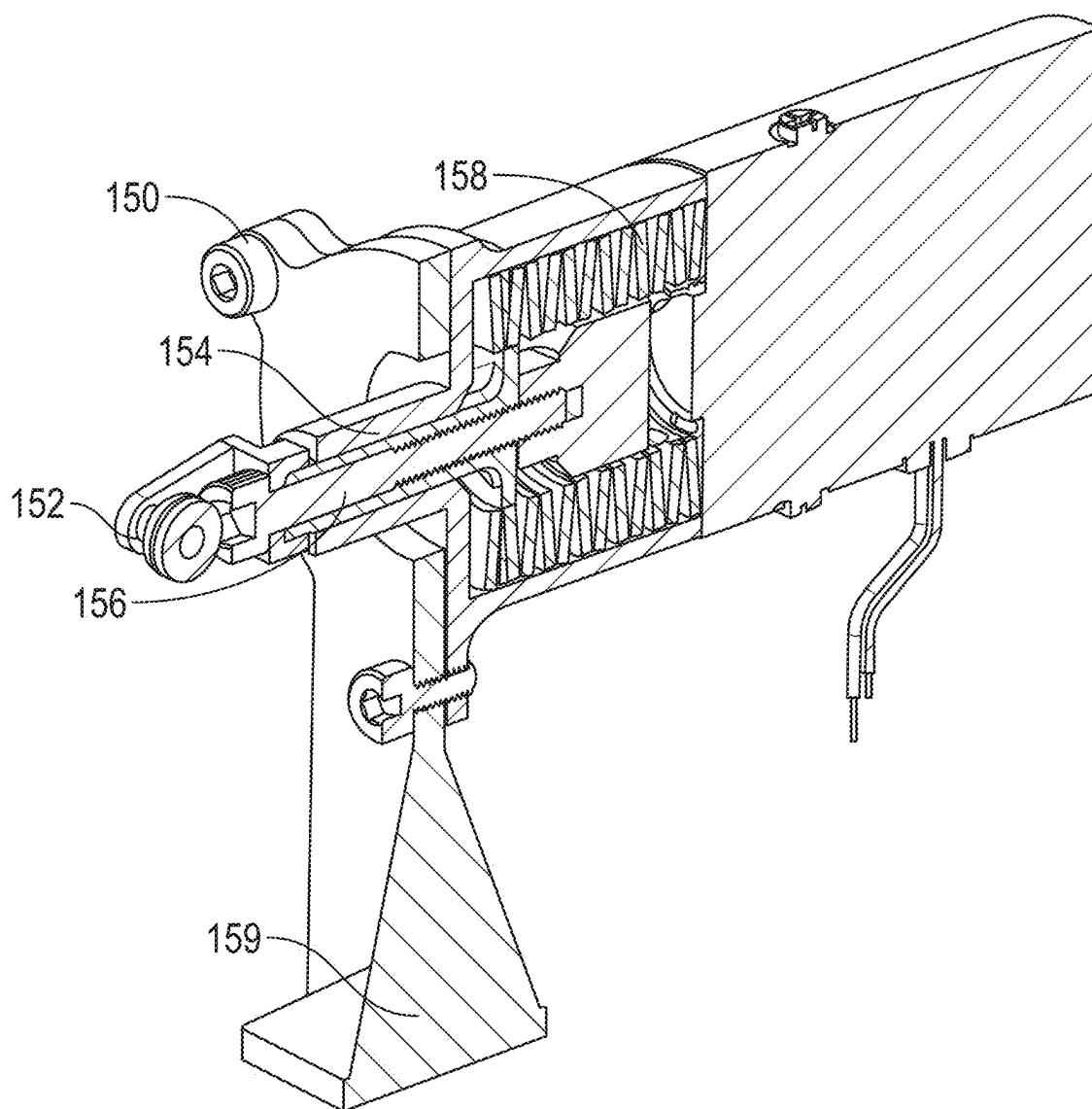


FIG. 8

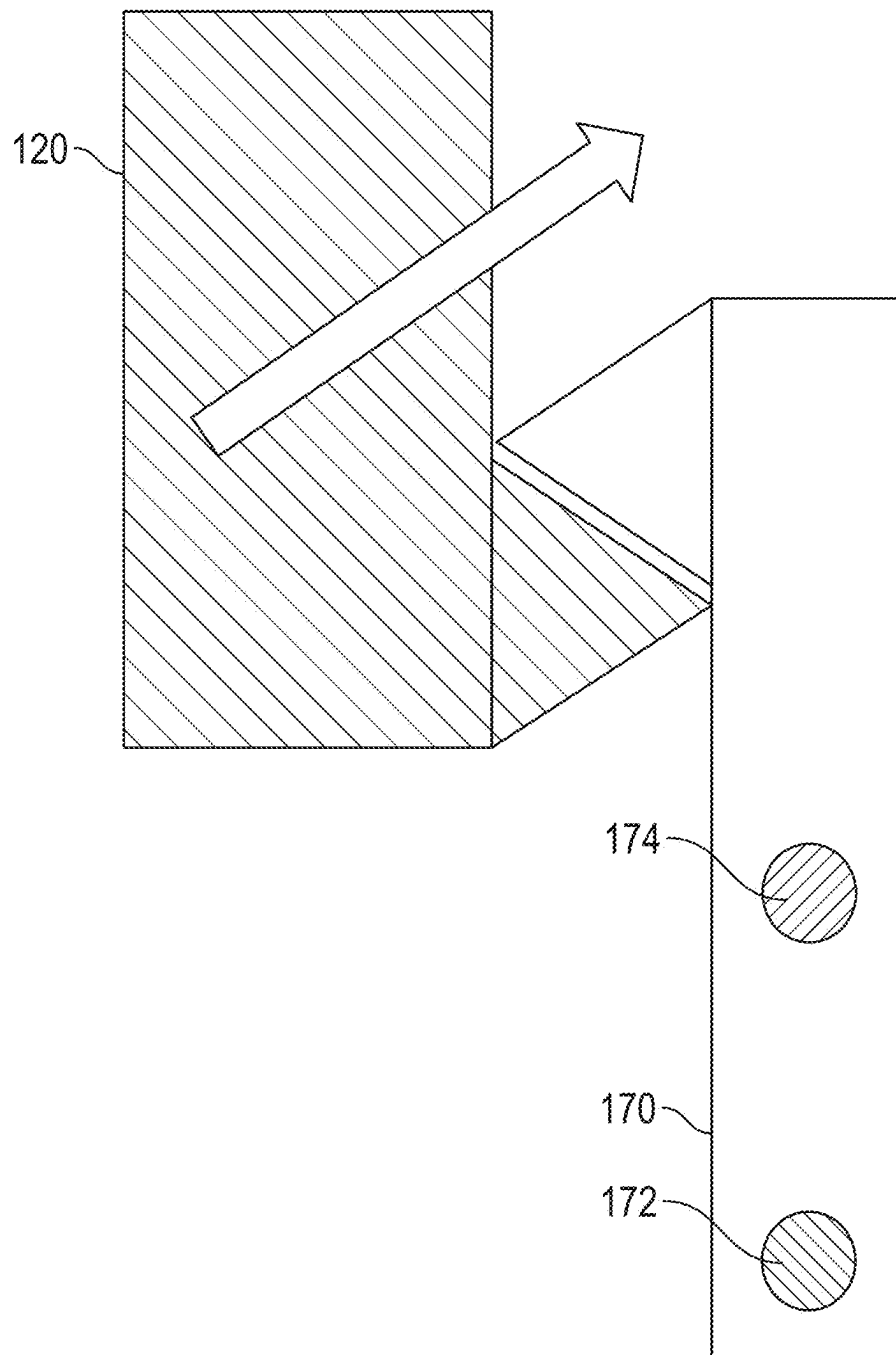


FIG. 9A

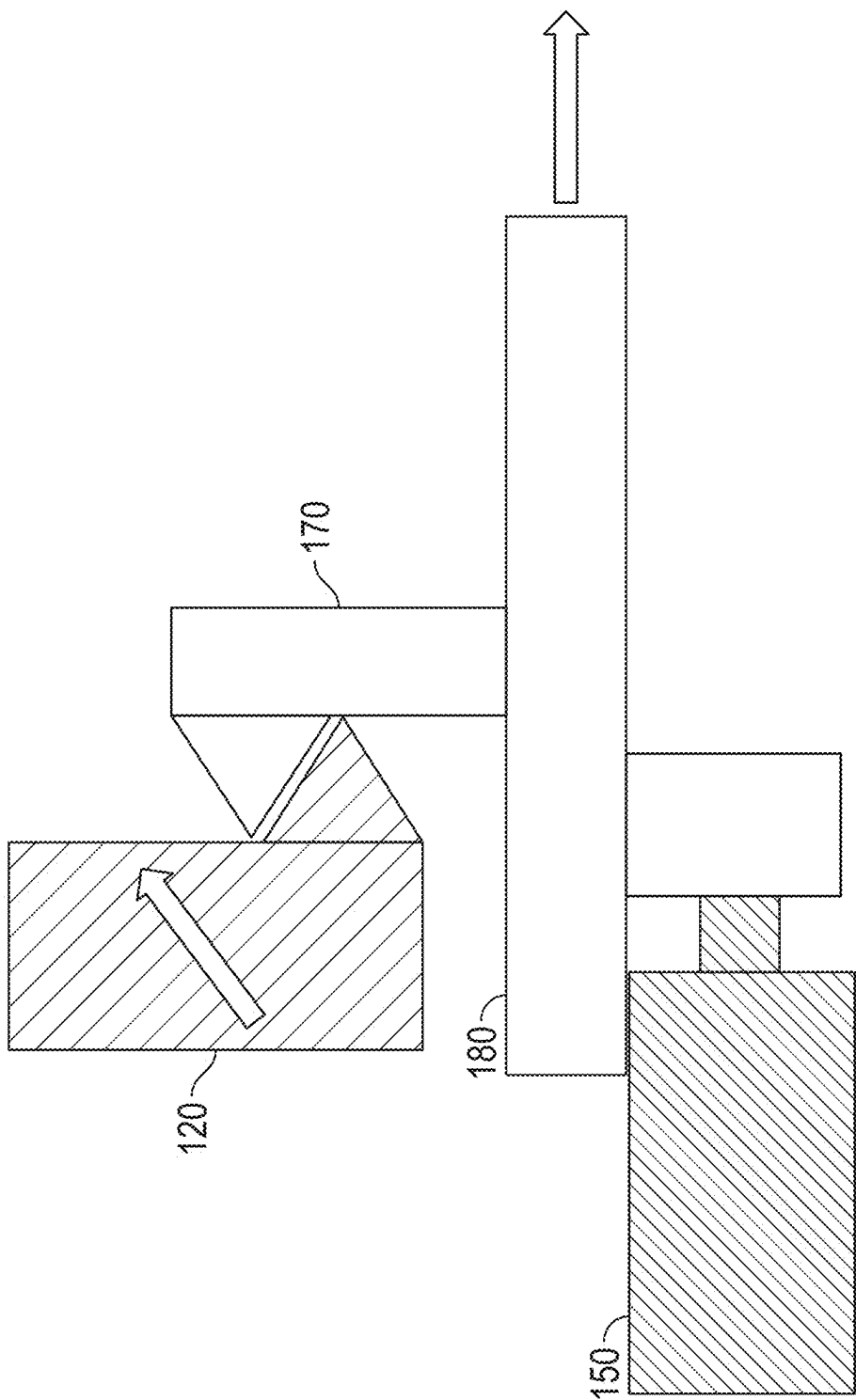


FIG. 9B

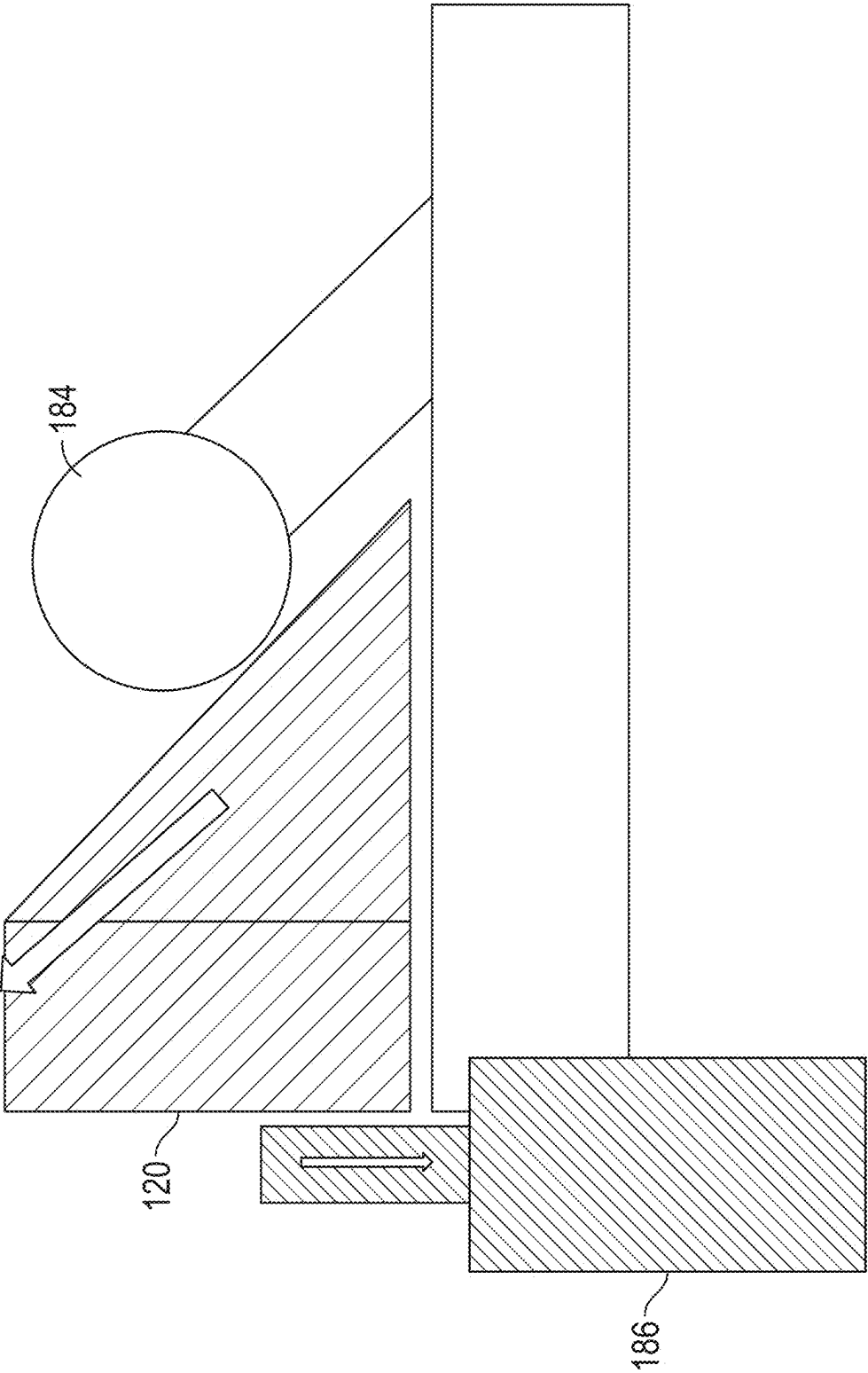


FIG. 10

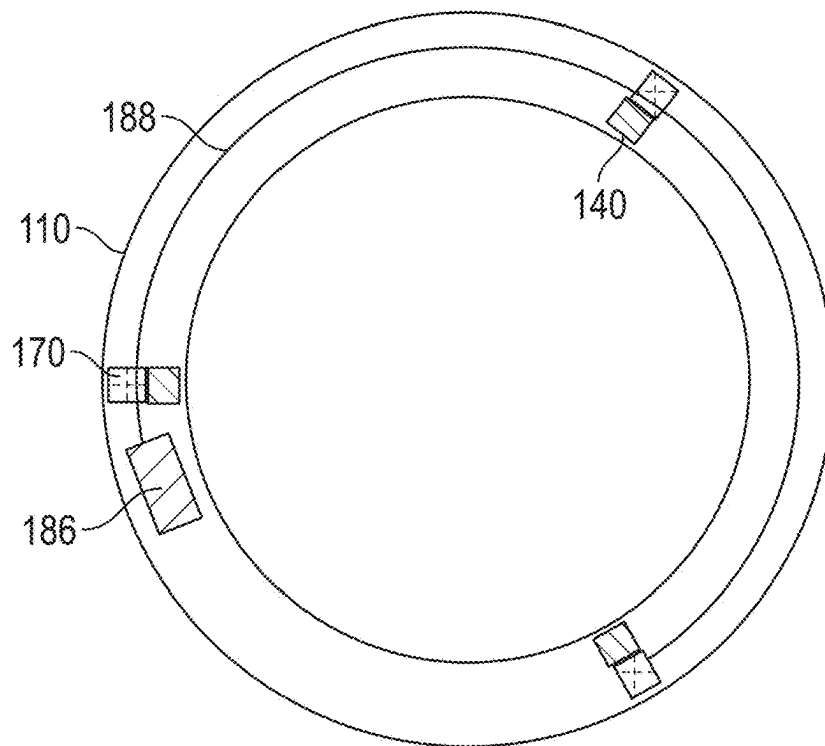


FIG. 11

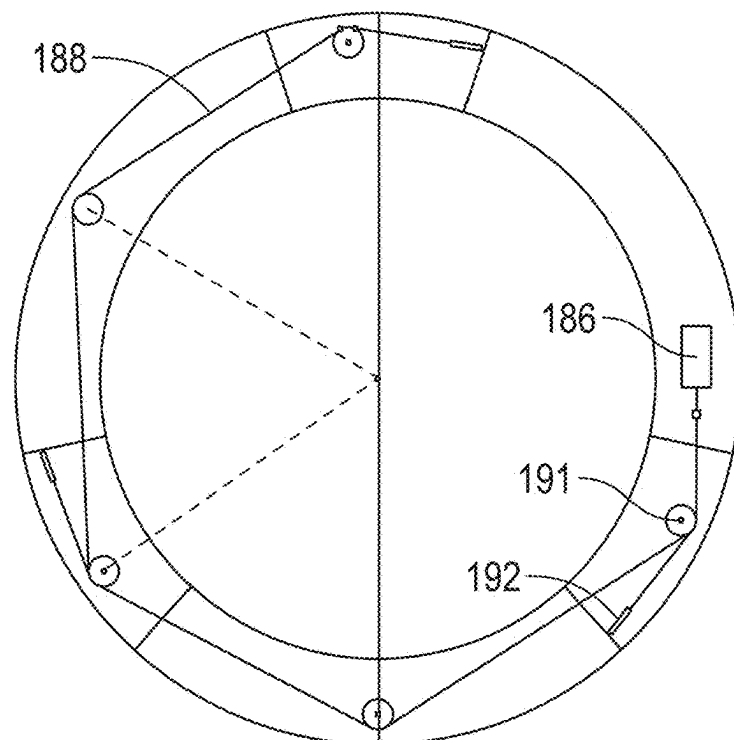


FIG. 12

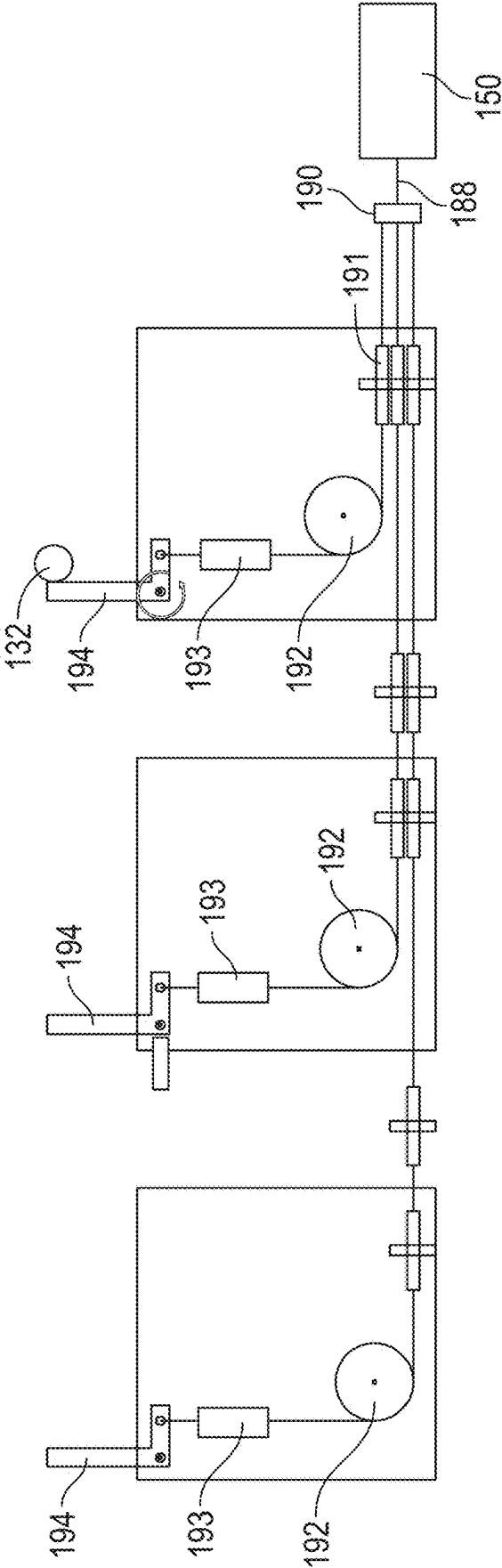


FIG. 13

1

SYSTEM AND METHOD OF EJECTING A PAYLOAD FROM AN EXTRATERRESTRIAL SURFACE

BACKGROUND OF THE DISCLOSURE

The subject matter disclosed herein relates to a system for ejecting a payload from an extraterrestrial vehicle that simultaneously translates and rotates the payload relative to the vehicle.

Marman clamps release rings are commonly used to deploy payloads from an extraterrestrial surface, such as on a spacecraft for example. A Marman clamp is a band clamp that allows two cylindrical interfaces to be clamped together. A typical Marman clamp system is found in NASA's Marman Clamp System Design Guidelines, Guideline No. GD-ED-2214.

Disclosed herein are payload ejection systems providing rotational force to a payload while reducing the amount of material needed for the systems as compared to existing solutions.

BRIEF DESCRIPTION OF THE DISCLOSURE

According to one aspect of the disclosure a spin ejection mechanism is disclosed including a first bracket; at least one second bracket spaced apart from the first bracket; at least one guide mechanism operably coupled between the first bracket and one of the at least one second brackets, the at least one guide mechanism configured to axially restrain the second bracket; at least one releasable lock mechanism operably coupled to the at least one second bracket and configured to initiate restrained movement of one of the at least one guide mechanism; and a spring-loaded lock mechanism coupled to the first bracket and configured to initiate unrestrained movement of the at least one guide mechanism.

In addition to one or more of the features described herein, or as an alternative, further embodiments of the mechanism may include a guide mechanism including at least one guide roller; and a helical guide rail engaged with the at least one guide roller, wherein the guide mechanism is configured to move the at least one guide roller along the helical guide rail upon release of the latch.

In addition to one or more of the features described herein, or as an alternative, further embodiments of the mechanism may include a helical guide rail including a cradle configured to retain the at least one guide roller in a second position.

In addition to one or more of the features described herein, or as an alternative, further embodiments of the mechanism may include a spring-loaded lock mechanism configured to move the at least one guide roller past the cradle.

In addition to one or more of the features described herein, or as an alternative, further embodiments of the mechanism may include at least one releasable lock mechanism configured to move the at least one guide roller into the cradle.

In addition to one or more of the features described herein, or as an alternative, further embodiments of the mechanism may include a spring-loaded lock mechanism arranged perpendicular to a plane of the first bracket.

In addition to one or more of the features described herein, or as an alternative, further embodiments of the mechanism may include each of the at least one releasable lock mechanisms configured to activate in series.

2

In addition to one or more of the features described herein, or as an alternative, further embodiments of the mechanism may include a spring assembly operably coupled between the first bracket and one of the at least one second bracket in a compressed position.

In addition to one or more of the features described herein, or as an alternative, further embodiments of the mechanism may include a spring assembly configured to move to an expanded position upon release of the spring-loaded lock mechanism.

In addition to one or more of the features described herein, or as an alternative, further embodiments of the mechanism may include a spring assembly is configured to propel the at least one second bracket as the spring assembly moves to the expanded position.

In addition to one or more of the features described herein, or as an alternative, further embodiments of the mechanism may include a spin ejection mechanism configured to launch an object situated on the at least one second bracket.

In addition to one or more of the features described herein, or as an alternative, a method of launching objects from extraterrestrial sites includes providing the spin ejection mechanism; coupling an object to the at least one second bracket; releasing the at least one releasable lock mechanism; and releasing the spring-loaded lock mechanism to initiate unrestrained movement of the at least one guide mechanism, moving the second bracket from a first position to a second position, the second position being in a direction away from the first bracket.

In addition to one or more of the features described herein, or as an alternative, the method includes moving at least one guide roller from a cradle along a helical guide rail of the guide mechanism upon release of the spring-loaded lock mechanism.

In addition to one or more of the features described herein, or as an alternative, the method includes selectively retaining the at least one roller in the cradle when in the second position.

In addition to one or more of the features described herein, or as an alternative, the method includes releasing the spring-loaded lock mechanism after the at least one guide roller is retained by the cradle.

In addition to one or more of the features described herein, or as an alternative, the method includes activating the at least one releasable lock mechanisms in series.

In addition to one or more of the features described herein, or as an alternative, the spin ejection mechanism further comprises a spring assembly operably coupled between the first bracket and one of the at least one second bracket in a compressed position.

In addition to one or more of the features described herein, or as an alternative, the method includes rotating the second bracket while moving the spring assembly to an expanded position upon release of the latch.

These and other advantages and features will become more apparent from the following description taken in conjunction with the drawings.

BRIEF DESCRIPTION OF DRAWINGS

The subject matter, which is regarded as the disclosure, is particularly pointed out and distinctly claimed in the claims at the conclusion of the specification. The foregoing and other features, and advantages of the disclosure are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:

3

FIG. 1 is a perspective view of a spin ejection mechanism supporting a payload in accordance with an embodiment;

FIG. 2A is a side view of the spin ejection mechanism of FIG. 1 depicting release of a releasable lock mechanism in accordance with an embodiment;

FIG. 2B is a side view of the spin ejection mechanism of FIG. 1 depicting a guide roller situated within a cradle in accordance with an embodiment;

FIG. 2C is a side view of the spin ejection mechanism of FIG. 1 depicting release of a spring-loaded lock mechanism in accordance with an embodiment;

FIG. 2D is a side view of the spin ejection mechanism of FIG. 1 depicting actuation of a guide roller by a spring-loaded lock mechanism in accordance with an embodiment;

FIG. 2E is a side view of the spin ejection mechanism of FIG. 1 depicting movement of a guide roller along a guide rail in accordance with an embodiment;

FIG. 2F is a side view of the spin ejection mechanism of FIG. 1 depicting deployment of a payload in accordance with an embodiment;

FIG. 3A is a section view of an undeployed releasable lock mechanism in accordance with an embodiment;

FIG. 3B is a section view of a deployed releasable lock mechanism in accordance with an embodiment;

FIG. 4 is a perspective view of the guide mechanism and the spring assembly of FIG. 1 in accordance with an embodiment;

FIG. 5 is a section view of a spring assembly of FIG. 1 in accordance with an embodiment;

FIG. 6 is a side view of the spring assembly of FIG. 1 in accordance with an embodiment;

FIG. 7 is a perspective view of the spring-loaded lock mechanism of FIG. 1 in accordance with an embodiment;

FIG. 8 is a section view of the spring-loaded lock mechanism of FIG. 1 in accordance with an embodiment;

FIG. 9A is a schematic depiction of a latch mechanism in accordance with an embodiment;

FIG. 9B is a schematic depiction of release of a latch mechanism in accordance with an embodiment;

FIG. 10 is a schematic depiction of a retention pin in accordance with an embodiment;

FIG. 11 is a top view of a spin ejection mechanism having a pin puller and connected latch mechanisms in accordance with an embodiment;

FIG. 12 is a top view of a spin ejection mechanism having a vertical pulley and pulley stack in accordance with an embodiment; and

FIG. 13 is a side view of the spin ejection mechanism of FIG. 12 in accordance with an embodiment.

The detailed description explains embodiments of the disclosure, together with advantages and features, by way of example with reference to the drawings.

DETAILED DESCRIPTION OF THE DISCLOSURE

One disadvantage of the Marman clamp is the mass and volume of the clamp required to deploy a payload. As the desire for larger and correspondingly heavier payloads increases, logistical issues are created in deploying the payloads from an extraterrestrial surface.

Other commonly used devices require precisely-timed simultaneous firing of release mechanisms to achieve a desired velocity and spin rate. Such devices may be prone to failure due to misfire or mistiming in one or more of the release mechanisms that causes an undesired force or torque loading on the payload.

4

While existing payload ejection systems are suitable for their intended purposes the need for improvement remains, particularly in providing a mechanism having the features described herein.

Embodiments of the present disclosure provide for precise deployment of a payload without requiring simultaneous release of multiple Hold Down Release Mechanisms (HDRMs). Further embodiments of the present disclosure provide for reduced or minimal tip-off and lateral velocity imparted on the payload. Further embodiments of the present disclosure impart low-uncertainty axial velocity and spin rate on the payload without initiating the simultaneous release of multiple Hold Down Release Mechanisms (HDRMs). The device does not use precise timing and “simultaneous” release of multiple release devices in order to provide predictable, reliable performance. Rather the device is configured to release one or more releasable lock mechanisms before release of a spring-loaded lock mechanism.

Referring now to FIG. 1, an embodiment is shown of a spin ejection mechanism 100. In an embodiment, the spin ejection mechanism 100 includes a first bracket 110 and a second bracket 120 spaced apart from the first bracket 110. The brackets 110, 120 may be mounted to a frame 105. In an embodiment, the frame 105 is integrated into a spacecraft or an extraterrestrial vehicle. The second bracket 120 may be separated from the first bracket 110 by a distance ranging from 10 cm-100 cm. In some embodiments, the distance between the first bracket 110 and the second bracket may be more than 100 cm. In some embodiments, the spin ejection mechanism 100 includes multiple first brackets 110 and multiple corresponding second brackets 120. FIG. 1 depicts an embodiment including three first brackets 110 and two second brackets 120 but embodiments of the spin ejection mechanism 100 may include different amount of brackets. It should be appreciated that in the illustrated embodiment the brackets 110, 120 are disposed equidistant from each other.

The spin ejection mechanism 100 may be configured to launch a payload 170 (i.e., an object, capsule, or sample) situated on the second bracket 120. As discussed in more detail herein, the mechanism 100 is configured to selectively release the second bracket 120 from the first bracket 110 (and hence the payload 170) with a desired translational and rotational motion.

The spin ejection mechanism 100 may further include at least one guide mechanism 130 operably coupled between a first bracket 110 and a second bracket 120. In some embodiments, the guide mechanism 130 is configured to directly axially restrain the second bracket 120. In embodiments with multiple first and second brackets, the spin ejection mechanism 100 includes a plurality of guide mechanisms 130 that are configured to selectively and axially restrain a respective second bracket 120. It should be appreciated that while the illustrated embodiments depict three guide mechanisms 130, this is for example purposes and the claims should not be so limited. In other embodiments, the spin ejection mechanism 100 may have more or fewer guide mechanisms 130.

The guide mechanism 130 may include at least one guide roller 132 and a helical guide rail 134 operably coupled to the guide roller 132. In the illustrated embodiment, the guide mechanism 130 includes a guide roller 132 extending from the second bracket 120 towards the first bracket 110. In some embodiments, the guide mechanism 130 include multiple guide rollers 132 that are spaced apart to allow the helical guide rail 134 to be disposed therebetween.

5

The helical guide rail **134** may include a cradle **136** (discussed in further detail below with regards to FIG. 2B) shaped to restrain movement of the guide roller **132**. The guide mechanism **130** may be configured to release the guide roller **132** from the cradle **136** upon release of a spring-loaded lock mechanism **150** as discussed in further detail below. In some embodiments, a patterned series of helical guide rails **134** and guide rollers **132** is provided and configured to control and constrain the motion of ejection of the payload **170** to a desired helix path (i.e., a combination of spin and axial velocity). The combination of helical guide rails **134** and guide rollers **132** may impart a reduced or minimal amount of tip-off and lateral velocity on the payload **170** while providing high precision for imparted axial velocity and spin rate.

At least one releasable lock mechanism **140** may be operably coupled to a first bracket **110** and operably coupled to a second bracket **120**. The releasable lock mechanism **140** may be configured to initiate restrained movement of the guide mechanism **130**. In some embodiments, the at least one releasable lock mechanism **140** is oriented to impart a force in a direction towards the second bracket **120**. Such a force causes movement of the second bracket **120**, for example in a counterclockwise direction. In an embodiment, the releasable lock mechanism **140** is a Model ERM2000 launch lock device manufactured by Ensign-Bickford Aerospace & Defense of Simsbury, Connecticut. The releasable lock mechanism **140** includes a biasing member (not shown), such as a spring for example, that is selectively released to initiate movement of the second bracket **120**.

In further embodiments, the releasable lock mechanisms **140** are configured to activate in series. In such embodiments, the order of activation of the releasable lock mechanisms **140** does not affect ejection of the payload **170**. No motion between the releasable lock mechanism **140** and payload **170** is expected while activation of the releasable lock mechanisms **140** occurs. The two steps of releasing the releasable lock mechanisms **140** before the spring-loaded lock mechanism **150** allows the payload **170** to be released without precise/simultaneous activation timing of the releasable lock mechanisms **140**.

The spin ejection mechanism **100** may further include a spring-loaded lock mechanism **150** coupled to a first bracket **110** and releasably coupled to the respective guide mechanism **130**. The spring-loaded lock mechanism **150** may be arranged substantially perpendicular to a plane of the first bracket **110**, imparting a force perpendicular to the plane of the first bracket **110**. In an embodiment, the spring-loaded lock mechanism **150** is a model ERM4000 launch lock device manufactured by Ensign-Bickford Aerospace & Defense of Simsbury, Connecticut.

The spin ejection mechanism **100** may further include at least one spring assembly **160** operably coupled between a first bracket **112** and a second bracket **120** when in a compressed position. Compression of spring assembly **160** is discussed in further detail below with regards to FIG. 6. The spring assembly **160** may be arranged on an angle relative to a plane of the first bracket **110**, the angle being greater than zero degrees and less than 90 degrees. In embodiments including multiple spring assemblies **160**, each spring assembly **160** may be oriented at the same angle. In an embodiment, the spring assembly **160** is arranged on an angle of 23 degrees relative to an axis extending perpendicular to the first bracket **110**.

Referring now to FIGS. 2A-2F, the operation of the mechanism **100** is shown and described. In this embodiment, the releasable lock mechanism **140** may be configured to

6

initiate restrained movement of the second bracket **120**. In embodiments with multiple second brackets **120**, each bracket **120** is paired with a corresponding releasable lock mechanism **140**. FIG. 2A depicts release of the releasable lock mechanism **140**, which is depicted in more detail in FIGS. 3A and 3B.

Releasable lock mechanism **140** may include an ejector release mechanism (ERM) **141** operably coupled to a coupler **143**. A portion of the ERM **141** and a portion of the coupler **143** may be surrounded by a cup **142**. Cup **142** may be fixedly couple to a first bracket **110**. A cone **144** may be situated above the cup **142** and may surround a portion of the coupler **143**. The cone **144** may be fixedly connected to a second bracket **120**. A fastener **145** may join the coupler **143** to a cone **144**. Release of the ERM **141** may release the coupler **143**, thereby releasing the cone **144** from the cup **142** and releasing the second bracket **120** from the first bracket **110**.

FIG. 2B depicts movement of the guide roller **132** towards a cradle **136**. Release of the releasable lock mechanism **140** initiates upward movement of the second bracket **120** and the guide roller **132** coupled to the second bracket **120**. In an embodiment a biasing member, such as spring assembly **160**, also contributes to the movement of the second bracket **120**. Movement of the guide roller **132**, and therefore the second bracket **120**, is restrained once the guide roller contacts the cradle **136**. FIG. 2C depicts the guide roller **132** sitting within the cradle **136**.

FIGS. 2C and 2D depicts release of the spring-loaded lock mechanism **150**. The spring-loaded lock mechanism **150** may be configured to cause the axial and rotational movement of the second bracket **120**. In some embodiments, the spring-loaded lock mechanism **150** provides sufficient force to displace the guide roller **132** from the cradle **136** in multiple guide mechanisms **130** of a spin ejection mechanism **100**. In an embodiment, the force applied by the spring loaded lock mechanism **150** is tangential to the mechanism **100**. The components of the spring-loaded lock mechanism **150** are described in further detail below with regards to FIGS. 7 and 8. FIG. 2E depicts movement of the guide roller **132** along the guide rail **134**. FIG. 2F depicts deployment of the second bracket **120**, propelling the payload **170**.

The components of the spin ejection mechanism **100** may be composed of materials suitable for use in outer space or on extraterrestrial sites. By way of non-limiting example, the lock mechanisms **140** and **150**, cup **142**, cone structure **144**, fastener **145**, guide roller **132**, and/or guide rails **124** may be composed of aluminum or stainless steel.

Referring now to FIGS. 4 and 5, each spring assembly **160** may include a body **161** having an end pivotally coupled to the first bracket **110**. For example, the end may be pivotally coupled by a cylindrical bearing hinged mounting point **166**. In some embodiments, each spring assembly **160** is connected to a second bracket **120** by an open ball and socket interface **165** on an end opposing the end pivotally coupled to the first bracket **110**. For example, the spring assembly **160** may include a ball joint **164** situated on an end releasably coupled to the socket **162** of the second bracket **120**. In some embodiments, disposed about the body **161** is a biasing member, such as compression spring **163**, that applies a force on the socket **162**.

The spring assembly **160** may be configured to move from a compressed state to an expanded position upon release of the releasable lock mechanism **140**. In some embodiments, the spring assembly **160** measures from 50 to 80 mm in length in a compressed state. In further embodiments, the spring assembly **160** measures from 60 to 70 mm in length

in a compressed state. In some embodiments, the spring assembly **160** measures from 100 to 150 mm in length an expanded position. In further embodiments, the spring assembly **160** measures from 125 to 140 mm in length at an expanded position.

Referring now to FIGS. 7 and 8, the spring-loaded lock mechanism **150** includes a kick-off roller **152** positioned to move the guide roller **132** out of the cradle **136**. At time of activation of the spring-loaded lock mechanism **150**, a biasing member, such as a stack of compressed washers **158** (for example, Belleville washers), within the mechanism **150** are released, extending a pushrod or piston **154** outwardly from the mechanism **150**. A preloaded fastener **156** may be housing within the pushrod/piston **154**. In some embodiments, the spring-loaded lock mechanism **150** is secured to a first bracket **110** by a mounting bracket **119**.

Referring now to FIGS. 9A to 13, an alternate embodiment is shown of the spin ejection mechanism **100** including latches and pulleys. For example, one or more latches **170** may hold at least one second bracket **120** in place. FIG. 11 depicts three latches with associated launch locks by way of non-limiting example. The latch **170** may include a release pin **172** situated at an end of the latch **170** proximate to the first bracket **110**. The latch **170** may further include a pivot **174** within the latch **170** positioned in a direction from the release pin **172** opposite the first bracket **110**. In some embodiments, the latch **170** is configured to release the second bracket **120** by moving about the pivot **174** radially outward from the second bracket **120**.

In some embodiments, releasable lock mechanisms **140** release and cause the second bracket **120** to start rotation and axial motion until the second bracket **120** becomes preloaded into one or more latches **170**.

FIG. 9B depicts an embodiment where a latch plate **180** is secured to each latch **170**. In such an embodiment, a spring-loaded lock mechanism **150** may be deployed to release the latch plate **180**, rotating the latch plate **180** and coupled latches **170** away from the second bracket **120** and allowing deployment of a payload.

Referring now to FIG. 10, an embodiment of the spin ejection mechanism **100** is shown that may include a retention pin **184** preventing rotation and axial motion of the second bracket **120**. In such an embodiment, releasable lock mechanisms **140** release and cause the second bracket **120** to start rotation and axial motion until the second bracket **120** becomes restrained by the retention pin **184**. The retention pin **184** may be retracted to allow rotation and axial motion of the second bracket **120**. The retention pin **184** may be retracted by a pin puller **186** operably connected to the retention pin **184**, for example by a cable.

Referring now to FIGS. 11-13, in some embodiments a pin puller **186** (or spring-loaded mechanism **150**) may pull on a cable or series of cables **188** that simultaneously releases each latch **170**. The cable **188** coupled to spring-loaded mechanism **150** may pass through a cable guide interface **190**. The cabling **188** may then pass through to a pulley stack **191**. Further cabling **188** may pass through to a vertical pulley **192**. The cabling **188** may continue to tensioner spring **193**. The cabling **188** may terminate at a retention tab **194**.

The retention tab **194** (FIG. 13) may, in a first position, restrain axial movement of a guide roller **132**. Release of the spring-loaded mechanism **150** results in release of the guide roller **132**, for example by tensioning the cable **188** through the cable guide interface **190**, pulley stack **191**, vertical pulley **192**, and tensioner spring **193** to rotate the retention tab **194** out of the path of the guide roller **132**.

Referring to FIG. 13, the spin ejection mechanism **100** may include a vertical pulley **192**, tensioner spring **193**, and retention tab **194** for each guide roller **132**. In embodiments with multiple guide rollers **132**, cabling **188** extending from the pulley stack **191** connected to the cable guide interface **190** may extend to other pulley stacks in series. The configuration of pulley stacks **191** allows a single spring-loaded lock mechanism **150** to actuate each pulley stack **191** at once.

It should be appreciated that while embodiments herein may refer to the use of the spin ejection mechanism with respect to a particular application, such as a spacecraft that is returning sample materials from an extraterrestrial location, such as the moon or lunar surface, this is for example purposes and the claims should not be so limited. In other embodiments, the spin ejection mechanism described herein may be used in connection with spacecraft originating from other terrestrial or extraterrestrial bodies, such as but not limited to Mars, asteroids, Kupier Belt objects, and Trans-Neptunian objects for example. In still further embodiments, the spin ejection mechanism may be used on moons/satellite objects of other solar system planets, such as Titan or Europa for example.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It should also be noted that the terms “first”, “second”, “third”, “upper”, “lower”, and the like may be used herein to modify various elements. These modifiers do not imply a spatial, sequential, or hierarchical order to the modified elements unless specifically stated.

Various embodiments of the invention are described herein with reference to the related drawings. Alternative embodiments of the invention can be devised without departing from the scope of this invention. Various connections and positional relationships (e.g., over, below, adjacent, etc.) are set forth between elements in the following description and in the drawings. These connections and/or positional relationships, unless specified otherwise, can be direct or indirect, and the present invention is not intended to be limiting in this respect. Accordingly, a coupling of entities can refer to either a direct or an indirect coupling, and a positional relationship between entities can be a direct or indirect positional relationship. Moreover, the various tasks and process steps described herein can be incorporated into a more comprehensive procedure or process having additional steps or functionality not described in detail herein.

The following definitions and abbreviations are to be used for the interpretation of the claims and the specification. As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” “contains” or “containing,” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a composition, a mixture, process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but can include other elements not expressly listed or inherent to such composition, mixture, process, method, article, or apparatus.

Additionally, the term “exemplary” is used herein to mean “serving as an example, instance or illustration.” Any embodiment or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. The terms “at least one” and “one or more” may be understood to include any integer

number greater than or equal to one, i.e. one, two, three, four, etc. The terms “a plurality” may be understood to include any integer number greater than or equal to two, i.e. two, three, four, five, etc. The term “connection” may include both an indirect “connection” and a direct “connection.”

The terms “about,” “substantially,” “approximately,” and variations thereof, are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” can include a range of +8% or 5%, or 2% of a given value.

For the sake of brevity, conventional techniques related to making and using aspects of the invention may or may not be described in detail herein. In particular, various aspects of computing systems and specific computer programs to implement the various technical features described herein are well known. Accordingly, in the interest of brevity, many conventional implementation details are only mentioned briefly herein or are omitted entirely without providing the well-known system and/or process details.

The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments described herein.

While the disclosure is provided in detail in connection with only a limited number of embodiments, it should be readily understood that the disclosure is not limited to such disclosed embodiments. Rather, the disclosure can be modified to incorporate any number of variations, alterations, substitutions or equivalent arrangements not heretofore described, but which are commensurate with the spirit and scope of the disclosure. Additionally, while various embodiments of the disclosure have been described, it is to be understood that the exemplary embodiment(s) may include only some of the described exemplary aspects. Accordingly, the disclosure is not to be seen as limited by the foregoing description, but is only limited by the scope of the appended claims.

What is claimed is:

1. A spin ejection mechanism comprising:
 - a first bracket;
 - at least one second bracket spaced apart from the first bracket;
 - at least one guide mechanism operably coupled between the first bracket and one of the at least one second brackets, the at least one guide mechanism configured to axially restrain the second bracket;
 - at least one releasable lock mechanism operably coupled to the at least one second bracket and configured to initiate restrained movement of one of the at least one guide mechanism; and
 - a spring-loaded lock mechanism coupled to the first bracket and configured to initiate unrestrained movement of the at least one guide mechanism.

2. The mechanism of claim 1, wherein the guide mechanism comprises:

- at least one guide roller; and
- a helical guide rail engaged with the at least one guide roller, wherein the at least one guide mechanism is configured to move the at least one guide roller along the helical guide rail upon release of a latch.

3. The mechanism of claim 2, wherein the helical guide rail comprises a cradle configured to retain the at least one guide roller in a second position.

4. The mechanism of claim 3, wherein the spring-loaded lock mechanism is configured to move the at least one guide roller past the cradle.

5. The mechanism of claim 3, wherein the at least one releasable lock mechanism is configured to move the at least one guide roller into the cradle.

6. The mechanism of claim 1, wherein the spring-loaded lock mechanism is arranged perpendicular to a plane of the first bracket.

7. The mechanism of claim 1, wherein each of the at least one releasable lock mechanisms are configured to activate in series.

8. The mechanism of claim 1, further comprising a spring assembly operably coupled between the first bracket and one of the at least one second bracket in a compressed position.

9. The mechanism of claim 8, wherein the spring assembly is configured to move to an expanded position upon release of the spring-loaded lock mechanism.

10. The mechanism of claim 9, wherein the spring assembly is configured to propel the at least one second bracket as the spring assembly moves to the expanded position.

11. The mechanism of claim 1, wherein the spin ejection mechanism is configured to launch an object situated on the at least one second bracket.

12. A method of launching objects from extraterrestrial sites, comprising:

- providing the spin ejection mechanism of claim 1;
- coupling an object to the at least one second bracket;
- releasing the at least one releasable lock mechanism; and
- releasing the spring-loaded lock mechanism to initiate unrestrained movement of the at least one guide mechanism, moving the second bracket from a first position to a second position, the second position being in a direction away from the first bracket.

13. The method of claim 12, further comprising moving at least one guide roller from a cradle along a helical guide rail of the at least one guide mechanism upon release of the spring-loaded lock mechanism.

14. The method of claim 13, further comprising selectively retaining the at least one roller in the cradle when in the second position.

15. The method of claim 14, further comprising releasing the spring-loaded lock mechanism after the at least one guide roller is retained by the cradle.

16. The method of claim 12, further comprising activating the at least one releasable lock mechanism in series.

17. The method of claim 12, wherein the spin ejection mechanism further comprises a spring assembly operably coupled between the first bracket and one of the at least one second bracket in a compressed position.

18. The method of claim 17, further comprising rotating the second bracket while moving the spring assembly to an expanded position upon release of a latch.

* * * * *